JARBY JESSID COSME

Software & AI Engineer | Production AI Systems, Backend & Data

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Available for full-time remote US nearshore roles (GMT-5, overlaps with EST/CST)

PROFESSIONAL SUMMARY

Software engineer with ~7 years of experience (Python, C#/.NET, TypeScript, SQL) who takes projects from zero to production. Sole technical owner of SIESA's AI support platform, with one goal: improve the service end to end, not "add a chatbot". Started from the data, built and run the RAG system that now answers the ERP support chat in every module, measured what was actually being resolved and orchestrated in LangGraph the agents that measure it. Presents business cases to the CEO. My public repos rebuild the evaluation method and the learning loop on open data.

PROFESSIONAL EXPERIENCE

Software Developer — SIESA, Cali, Colombia
Mar 2022 – Present

AI support platform — sole technical owner (2025 – Present)

- **Diagnosis first:** analyzed 90 days of production conversations and presented the business case for an in-house engine to leadership, with 6 proposals ranked by impact and cost.
- **A RAG built for the service:** designed, built and operate the RAG platform behind the support chat, on three rules: never make things up (citation and numeric validation, deterministic guards), hand off to a person on low confidence (tool calling) and never brush the customer off. Human escalation per chat in Payroll dropped, confirmed by a significance test, and replies that send customers to "contact support" became rare. *Framework-free Python on the critical path; managed LLMs split between a large and a small model; vector search with reranking.*
- **Quality measured, not assumed:** golden sets with an LLM judge gating releases, A/B replay on real traces and significance tests. Answer correctness rose between evaluation cuts (human-validated judge) and prompt caching lowered the cost per turn. Measured hybrid BM25 + RRF retrieval twice and retired it: the lever was the content (reprocessed the document base with multimodal OCR).
- **Measuring what gets resolved — Jarvis:** built Jarvis, a daily LLM judge that decides who resolved each conversation (AI, agent or nobody) with the same rule for both. It grew out of the judge I already ran, replaced the previous ticket classifier and re-types every ticket at close, in production.
- **LangGraph orchestration:** implemented in LangGraph and LangChain Jarvis's closing engine and the service's nightly processes (resolution judge, scoring, ticket typing and expert-approved curation): typed state, parallel judges, PostgreSQL checkpoints to resume a run and human-review interrupts. Rewrote the bot pipeline as a graph with parity proven on its full test suite.
- **Built for others to build on:** a self-improvement loop with one-click expert approval that adds new Q&As every week at a minimal model cost and helped lower Commercial escalation, an admin console with per-module permissions and a model gateway other teams use. Ran production in the cloud with per-stage observability (p50/p95 latency, cost per turn) and module-by-module releases with rollback plans.
- **The service, not just the model:** redesigned support operations (per-agent WIP limits, FIFO assignment, load balancing). Customer satisfaction rose clearly in the weekly survey, and first response time and peak load per agent went down.
- **What's next:** designing an online triage channel that will route each request to AI, L1, L2 or L3 (its ceiling estimated from data first) and building RAG v2, which will act on customer systems (restarts, updates) with per-action permission levels. Trained 16 people across 3 teams on AI adoption.

Internal AI assistant & automation (2024 – 2025)

- Built the internal AI assistant (LLM API with tool calling over internal databases, the in-house Jira-style platform's requirements and support tickets): support agents, consultants and operations teams asked in natural language instead of opening several systems. Orchestration prototyped in LangChain; the production path went directly against the API.
- Automated the operational loop with n8n (alerts to agents on tickets close to their SLA, TTR and FCR targets) and Power BI dashboards fed from the Jira-style platform.

Data, automation and platforms (Mar 2022 – Present)

- Replaced manual reporting with a Python pipeline (ticketing API → SQL Server with transactional loads and hash integrity checks → Power BI via REST API; validated against the manual load before replacing it) and build medallion ETL pipelines (GitHub → SQL Server); cut a daily SQL Server load time by more than half with identical results.
- Lead the in-house service and project management platform (Jira-style) used across the company; integrated 3 applications. Migrated 3 monoliths to microservices (.NET 8 / ASP.NET Core) in an agile team and modernized React/Angular modules.

Independent AI Consultant & Instructor — Part-time, remote

2026 – Present

- Founded and build Tocaturno (tocaturno.com), a production SaaS for 24/7 WhatsApp reception (bookings, reminders and handoff to a person for sensitive cases). Design AI agents and automations for clients (WhatsApp, n8n, MCP servers) and teach hands-on AI classes.

Freelance Software Developer — Self-employed, remote

Jun 2019 – Feb 2022

- Built 6+ full-stack web applications for Colombian SMBs (.NET Core, React, JavaScript; 1,500+ end users), including a PayU e-commerce platform (US$25K+ in first-year sales) and a multi-site inventory system (stock discrepancies –90%).

TECHNICAL SKILLS

- **AI & LLM:** RAG • AI Agents • LangGraph / LangChain (state graphs, checkpoints, human-in-the-loop) • Tool Calling • MCP • Embeddings & Semantic Search (pgvector) • Reranking • LLM Evaluation (LLM-as-Judge, golden sets, A/B) • LLMOps • Guardrails & Answer Validation • Multimodal OCR • Claude (AWS Bedrock) • Azure OpenAI / AI Foundry • OpenAI API • n8n
- **Backend & Frontend:** Python • FastAPI • C# • .NET 8 • ASP.NET Core • REST APIs • WebSockets • Microservices • System Design • Clean Architecture / SOLID • Testing (pytest, xUnit/NUnit) • React • TypeScript • Angular
- **Data & Cloud:** SQL Server (T-SQL) • PostgreSQL / pgvector • ETL Pipelines • Medallion Architecture • Power BI • AWS (Bedrock, EC2, S3, IAM) • Azure DevOps • GCP (Cloud Run) • Linux • Docker • CI/CD • Git
- **Practices & Leadership:** End-to-end ownership • Measured releases (baseline, A/B, rollback) • Observability (traces, latency, cost) • Security (IAM, secrets, JWT, RBAC) • AI-assisted development (Claude Code, Cursor) • Stakeholder communication & executive reporting • Technical mentoring

EDUCATION

- **B.Sc. in Systems Engineering** — UNAD — 2024 – Present (expected 2027) • **Technologist in Systems Analysis and Development** — SENA — 2020 – 2022